SVKM's NMIMS

Mukesh Patel School of Technology Management & Engineering

MANAGING UNCERTAINTY

Unit-wise Formula and Decision Guide

MBA Tech Data Science | Semester III | 2026–27

Black-and-white examination reference

What this guide does

- Covers the seven syllabus units, including distributions and statistical inference.
- Boxes each key formula and explains its symbols immediately below it.
- Shows exactly how to translate "less than", "at least", and other probability statements.
- Uses *test statistic versus critical value* only, not p-values.
- Distinguishes sample and population formulas, one-tail and two-tail tests, and equal versus unequal variance cases.

Sources. The syllabus excerpts supplied with the request and *Managing Uncertainty: Year-wise Worked Examination Papers, 2022–23 to 2025–26* (105-page compiled PDF):

`https://aam-007.github.io/vault/sem_3/pyqs/mu/TEE_Managing_Uncertainty_Solutions.pdf`

Independently written formula notes; not a reproduction of the past-paper answers.

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Start here: read the problem correctly

First identify what is being measured. A number of defects is a *count*; a waiting time is a *continuous measurement*. The same inequality words can lead to different endpoints in those two cases.

Probability language for an integer-valued count X

Words in question	Write and calculate
Exactly k	$P(X = k)$
Less than k; fewer than k	$P(X < k) = P(X \leq k - 1)$
At most k; no more than k	$P(X \leq k)$
More than k	$P(X > k) = 1 - P(X \leq k)$
At least k; k or more	$P(X \geq k) = 1 - P(X \leq k - 1)$
From a through b, inclusive	$P(a \leq X \leq b) = P(X \leq b) - P(X \leq a - 1)$
Between a and b, strict	$P(a < X < b) = P(X \leq b - 1) - P(X \leq a)$

Meaning. k, a, b are integers; P means probability. The $a - 1$ and $b - 1$ adjustments apply only because X is an integer count.

Probability language for a continuous variable X

$P(X < a) = P(X \leq a) = F_X(a)$
$P(X > a) = P(X \geq a) = 1 - F_X(a)$
$P(a < X < b) = P(a \leq X \leq b) = F_X(b) - F_X(a)$

Meaning. a, b are real-number cutoffs with $a < b$; $F_X(c) = P(X \leq c)$ is the cumulative distribution function (CDF). For a continuous variable, $P(X = c) = 0$.

Exam rule. For *exactly two* defects use $P(X = 2)$. For *fewer than two* defects use $P(X = 0) + P(X = 1)$. For *at least two* use $1 - P(X = 0) - P(X = 1)$. Write the allowed values before calculating.

Model-selection guide

Question describes	Start with
One success / failure	Bernoulli
Successes in a fixed number of independent trials	Binomial
Rare events counted in a fixed interval	Poisson
Trial number of first success	Geometric
Equal chance on a continuous interval	Continuous uniform
Time until the next constant-rate event	Exponential
Bell-shaped measurements; mean and SD given	Normal
Probability after learning new information	Conditional probability / Bayes
Two variables together; joint table	Joint PMF; marginal and conditional probabilities
Claim about one or two means	z-test / t-test depending on SD and assumptions

1. Unit 1 — Classification, tabulation and sampling

Core idea. Classification groups observations; tabulation displays the groups and counts. This unit is mainly definitions, but its short formulas support frequency-table questions.

Four classifications

Qualitative	Categories: department, colour, satisfaction level
Quantitative	Numbers: age, marks, defects
Chronological	Grouped by time: month or year
Geographical	Grouped by place: city or state

Frequency, total and relative frequency

$$N = \sum f_i, \qquad r_i = \frac{f_i}{N}, \qquad \% = 100 r_i$$

Symbols. f_i: count in group i; N: total count; r_i: proportion in that group; i: group number. All r_i add to 1.

Class midpoint, width and cumulative frequency

$$m_i = \frac{L_i + U_i}{2}, \quad h_i = U_i - L_i, \quad C_i = \sum_{j=1}^{i} f_j$$

Symbols. L_i, U_i: lower and upper *class boundaries*; m_i: midpoint; h_i: class width; C_i: cumulative frequency up through class i; j: class counter.

Recover a class frequency from cumulative data

$$f_i = C_i - C_{i-1}$$

Symbols. f_i: count in the current class; C_i: current cumulative count; C_{i-1}: preceding cumulative count, with $C_0 = 0$.

Sampling fraction

$$f = \frac{n}{N}$$

Symbols. n: units selected for the sample; N: units in the population; f: sampling fraction. Here f is *not* a class frequency.

Sampling designs. Simple random: every eligible unit has a known selection chance. Systematic: select every k-th unit after a random start. Stratified: sample within each defined group. Cluster: sample whole selected groups. Convenience: select easy-to-reach observations, which can produce bias.

Systematic selection step (when appropriate)

$$k \approx \frac{N}{n}$$

Symbols. N: size of ordered sampling list; n: desired sample size; k: approximate interval between selected units; use a random starting point and a valid design.

Exam rule. Do not mistake N (the population or total frequency) for n (the sample size). In grouped tables, state clearly what your total represents.

2. Unit 2 — Descriptive statistics and graphs

2.1 Mean, median and mode

Arithmetic mean (raw values)

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Symbols. x_i: observation i; n: number of observations; $\bar{x}$: sample mean. For a full population, the same calculation is often denoted μ and uses population size N.

Mean from a frequency table

$$\bar{x} \approx \frac{\sum f_i m_i}{N}, \qquad N = \sum f_i$$

Symbols. f_i: class frequency; m_i: class midpoint or listed discrete value; N: total frequency. For grouped intervals, this is an *estimate*.

Weighted mean

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Symbols. x_i: value or group mean; w_i: its weight, such as group size; $\bar{x}_w$: weighted mean. Do not give unequal groups equal weight.

Assumed-mean (coded) grouped mean

$$\bar{x} \approx A + h\,\frac{\sum f_i u_i}{N}, \quad u_i = \frac{m_i - A}{h}$$

Symbols. A: chosen reference midpoint; h: common midpoint spacing; m_i: midpoint; u_i: coded value; f_i: frequency; $N = \sum f_i$.

Median of ungrouped data — sort first.
For odd n: middle observation at position $(n + 1)/2$.
For even n: average the observations at positions $n/2$ and $n/2 + 1$.

Meaning. n: number of sorted observations; "position" starts at 1. These rules work for ungrouped data.

Median for grouped continuous classes

$$\text{Median} \approx L + \frac{N/2 - C}{f_m}\,h$$

Symbols. L: lower boundary of *median class*; N: total frequency; C: cumulative frequency *before* that class; f_m: frequency of median class; h: class width. Median class is the first whose cumulative frequency reaches or exceeds $N/2$.

Grouped mode

$$\text{Mode} \approx L + \frac{d_1}{d_1 + d_2}\,h$$

Symbols. L: lower boundary of modal class; h: its width; $d_1 = f_m - f_-$; $d_2 = f_m - f_+$; f_m: largest class frequency; f_-, f_+: preceding and following frequencies.

Exam rule. The *modal class* is the class with the largest frequency. The *mode* is the estimated value inside that class. Do not replace it with the class midpoint when the grouped-mode formula is required.

2.2 Quartiles, percentiles and spread

Grouped first and third quartiles

$$Q_1 \approx L_1 + \frac{N/4 - C_1}{f_1}\, h_1, \quad Q_3 \approx L_3 + \frac{3N/4 - C_3}{f_3}\, h_3$$

Symbols. Q_1, Q_3: 25th and 75th percentiles; for each target class, L is its lower boundary, C is the cumulative count before it, f is its frequency, h is its width; subscripts 1 and 3 refer to the classes containing positions $N/4$ and $3N/4$.

Interquartile range, quartile deviation

$$\text{IQR} = Q_3 - Q_1, \qquad \text{QD} = \frac{Q_3 - Q_1}{2}$$

Symbols. Q_1, Q_3: first and third quartiles; IQR: middle-half spread; QD: semi-interquartile range.

Grouped percentile (general form)

$$P_k \approx L + \frac{kN/100 - C}{f}\, h$$

Symbols. P_k: k-th percentile; k: percentile number from 0 to 100; L, C, f, h: lower boundary, preceding cumulative frequency, class frequency and width of the class containing position $kN/100$; N: total count.

Exam rule. For *ungrouped* quartiles, textbooks use different position and interpolation conventions. Follow the convention in your class notes. The grouped interpolation formulas above apply to continuous class intervals.

2.3 Visualisation: what to place on each axis

Chart	What it shows
Bar chart	Categories on horizontal axis; separate bars for counts/percentages
Pie chart	Categories as shares of a whole
Histogram	Continuous class intervals; adjoining rectangles; *area* represents frequency
Frequency polygon	Class midpoints joined using frequency heights (equal-width case)
Less-than ogive	Upper class boundaries versus cumulative frequency
More-than ogive	Lower class boundaries versus remaining cumulative frequency
Scatter plot	Paired (x, y) measurements; shows association

Pie-chart sector angle

$$\theta_i = 360^\circ \frac{f_i}{N}$$

Symbols. θ_i: sector angle in degrees; f_i: category count; N: total count.

Histogram frequency density (unequal widths)

$$\text{height}_i = \frac{f_i}{h_i}$$

Symbols. f_i: class frequency; h_i: class width; height is frequency density so rectangle area equals frequency. With equal widths, using frequencies directly gives proportional areas.

Exam rule. For classes 20–24, 25–29, etc. of whole-number measurements, use continuous class boundaries 19.5–24.5, 24.5–29.5, etc. before drawing a histogram.

3. Unit 3 — Variability, skewness and correlation

3.1 Population, sample and grouped-data spread

Range

$$R = x_{\max} - x_{\min}$$

Symbols. $x_{\max}$: largest observation; $x_{\min}$: smallest observation; R: range.

Population variance and standard deviation

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}, \qquad \sigma = \sqrt{\sigma^2}$$

Symbols. x_i: population value; μ: population mean; N: population size; σ^2: population variance; σ: population standard deviation.

Sample variance and standard deviation

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}, \qquad s = \sqrt{s^2}$$

Symbols. x_i: sample value; $\bar{x}$: sample mean; n: sample size, at least 2; s^2: sample variance; s: sample standard deviation. The divisor $n - 1$ estimates population variance.

Quick raw-data variance calculations

$$s^2 = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n - 1}, \quad \sigma^2 = \frac{\sum x_i^2}{N} - \mu^2$$

Symbols. x_i: values; n: sample size for s^2; N: population size for σ^2; μ: population mean. Use the correct formula for what the data represent.

Grouped descriptive variance

$$\sigma_g^2 \approx \frac{\sum f_i m_i^2}{N} - \bar{x}^2, \quad \sigma_g = \sqrt{\sigma_g^2}$$

Symbols. f_i: group frequency; m_i: group midpoint; $N = \sum f_i$; $\bar{x} = \sum f_i m_i / N$; σ_g^2: descriptive variance of midpoint-approximated data.

Grouped sample variance (if estimating a population)

$$s_g^2 \approx \frac{\sum f_i (m_i - \bar{x})^2}{N - 1}$$

Symbols. f_i: class count; m_i: midpoint; $\bar{x}$: grouped mean; $N = \sum f_i$ is now the *sample* count; s_g^2: midpoint-based sample variance estimate.

Coefficient of variation

$$\mathrm{CV} = \frac{s}{\bar{x}} \times 100\%$$

Symbols. s: sample standard deviation; $\bar{x}$: positive sample mean; CV: relative spread in percent. For a population use σ/μ. Most useful for comparing positive-scale data with meaningful zero.

Chebyshev's inequality

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \quad (k > 1)$$

Symbols. X: random variable with finite variance; μ: mean; σ: SD; k: positive number of SDs, here greater than 1. Valid for any distribution with finite variance. The bound is for *strictly within* $k\sigma$.

Meaning. Example: $k = 2$ guarantees at least 75% strictly within two standard deviations.

3.2 Shape: skewness and kurtosis

Karl Pearson's skewness

$$\mathrm{Sk} = \frac{\bar{x} - \mathrm{Mode}}{s}$$

Symbols. $\bar{x}$: mean; Mode: mode; s: sample SD (or use σ for a descriptive population measure); Sk: skewness coefficient. Positive: longer right tail. Negative: longer left tail.

Pearson's alternate skewness (when mode is unclear)

$$\mathrm{Sk}_2 = \frac{3(\bar{x} - \mathrm{Median})}{s}$$

Symbols. $\bar{x}$: mean; Median: middle value; s: sample SD; Sk_2: alternative approximate skewness coefficient.

Bowley's quartile skewness

$$\mathrm{Sk}_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

Symbols. Q_1, Q_2, Q_3: first quartile, median and third quartile; requires $Q_3 > Q_1$.

Moment-based kurtosis and excess

$$\beta_2 = \frac{m_4}{m_2^2}, \qquad \gamma_2 = \beta_2 - 3$$

Symbols. $m_2 = \sum(x_i - \bar{x})^2/n$: second central moment; $m_4 = \sum(x_i - \bar{x})^4/n$: fourth central moment; n: number of values; β_2: kurtosis; γ_2: excess kurtosis. Normal distribution has population $\gamma_2 = 0$.

Meaning. Positive excess: heavier tails than a normal distribution in the kurtosis sense; negative excess: lighter tails. Different software uses different sample corrections.

3.3 Covariance, Pearson and Spearman correlation

Sample covariance

$$s_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Symbols. (x_i, y_i): paired observations; $\bar{x}, \bar{y}$: their sample means; n: number of pairs; s_{xy}: sample covariance. Positive means values tend to move together.

Pearson correlation: simple idea

$$r = \frac{s_{xy}}{s_x s_y}$$

Symbols. s_{xy}: sample covariance; s_x, s_y: sample standard deviations of X, Y; r: Pearson correlation between -1 and 1. Both SDs must be nonzero.

Pearson correlation: direct calculation table

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{\left[n\sum x^2 - (\sum x)^2\right]\left[n\sum y^2 - (\sum y)^2\right]}}$$

Symbols. n: number of paired rows; $\sum x, \sum y$: column totals; $\sum xy$: sum of rowwise products; $\sum x^2, \sum y^2$: sums of squared values; r: Pearson correlation.

Spearman rank correlation: no tied ranks

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

Symbols. n: number of pairs; $d_i = R_i - S_i$: difference between the two ranks of pair i; R_i, S_i: ranks of x_i, y_i; r_s: Spearman correlation. Valid *only if there are no ties*.

If any values tie. Give tied values their *average rank*; then calculate ordinary Pearson correlation on the two columns of ranks. Do *not* blindly use the no-ties shortcut above.

Choose Pearson versus Spearman. Pearson uses paired numerical values and measures *linear* association; Spearman uses ranks and measures *monotonic* association. Even if the data are numerical marks, a question can explicitly ask for Spearman: rank both columns first. Correlation alone does not show causation.

Exam rule. For Pearson, your table columns are x, y, xy, x^2, y^2. For Spearman without ties, they are x, y, R, S, d, d^2. Rank both variables in the same direction (ascending or descending).

4. Unit 4 — Probability, independence and distributions

4.1 Basic probability and conditional probability

Classical probability: equally likely outcomes

$$P(A) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

Symbols. A: event of interest; all outcomes counted in the denominator must be equally likely.

Experimental (frequentist) probability

$$P(A) \approx \frac{\text{times } A \text{ occurs}}{\text{number of trials}}$$

Symbols. A: event of interest; the ratio estimates a long-run event probability from repeated comparable trials.

Complement, union and intersection

$$P(A^c) = 1 - P(A), \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Symbols. A, B: events; A^c: not A; $A \cup B$: A or B or both; $A \cap B$: both A and B. If mutually exclusive, $P(A \cap B) = 0$.

Conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Symbols. A: event sought; B: information already known; $P(B) > 0$. The vertical bar means *given*.

Multiplication rule

$$P(A \cap B) = P(B)P(A \mid B)$$

Symbols. A, B: events; $P(B)$: chance of B; $P(A \mid B)$: chance of A once B has occurred. Equivalently use $P(A)P(B \mid A)$.

Independent events

$$P(A \cap B) = P(A)P(B)$$

Symbols. A, B: events; equality defines independence. For $P(B) > 0$, it also means $P(A \mid B) = P(A)$. Independence is *not* the same as mutual exclusion.

Law of total probability (two sources)

$$P(E) = P(E \mid A)P(A) + P(E \mid A^c)P(A^c)$$

Symbols. E: observed event; A, A^c: two non-overlapping and exhaustive sources; each $P(E \mid \cdot)$ is the chance of E within that source.

Bayes' theorem: easiest two-source version

$$P(A \mid E) = \frac{P(E \mid A)P(A)}{P(E)}$$

Symbols. A: source or cause; E: observed evidence; $P(A)$: prior; $P(E \mid A)$: likelihood; $P(E)$: overall probability of evidence from total probability; $P(A \mid E)$: posterior.

Bayes' theorem: several possible sources

$$P(B_j \mid E) = \frac{P(B_j)P(E \mid B_j)}{\sum_i P(B_i)P(E \mid B_i)}$$

Symbols. B_i: mutually exclusive, exhaustive possible sources; j: desired source index; i: summation index; E: observed event; denominator $P(E) > 0$.

Exactly one of two events

$$P(A \triangle B) = P(A) + P(B) - 2P(A \cap B)$$

Symbols. $A \triangle B$: A or B, but not both; $A \cap B$: both events.

Without replacement: hypergeometric count

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

Symbols. N: population items; K: items of desired type; n: items drawn *without replacement*; X: desired-type count; k: specified count; $\binom{a}{b}$: combinations. Use only possible k values.

Exam rule. For *with replacement*, successive draws can be independent. For *without replacement*, the second draw's chances usually change. Never apply the binomial model automatically to draws without replacement.

4.2 Discrete distribution model sheet

A probability mass function (PMF) gives $P(X = k)$ for an *integer or other discrete value* k. Its probabilities are nonnegative and add to 1. For a count, $k = 0, 1, 2, \ldots$ as permitted by the model.

Bernoulli: one success/failure trial

$$P(X = x) = p^x(1-p)^{1-x}, \quad x = 0, 1$$

Symbols. X: indicator (1=success, 0=failure); p: success chance with $0 \le p \le 1$; x: realised outcome. $E(X) = p$; $\mathrm{Var}(X) = p(1-p)$.

Binomial: successes in n trials

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Symbols. X: total successes; n: fixed number of independent trials; k: successes, $0 \le k \le n$; p: constant success chance; $\binom{n}{k} = n!/[k!(n-k)!]$; !: factorial, including $0! = 1$.

Binomial mean, variance and SD

$$\mu = np, \quad \sigma^2 = np(1-p), \quad \sigma = \sqrt{np(1-p)}$$

Symbols. n: trial count; p: success probability; μ: expected successes; σ^2: variance; σ: SD.

Binomial checklist. Fixed n, two possible results per trial, same p, independent trials. If any key condition fails, reconsider the model.

Poisson: event count in a fixed interval

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Symbols. X: number of events; $k = 0, 1, 2, \ldots$; $\lambda > 0$: expected event count *for the stated interval*; $e \approx 2.71828$: exponential constant; !: factorial.

Poisson mean, variance and rate conversion

$$\mu = \sigma^2 = \lambda, \qquad \lambda = rt$$

Symbols. λ: expected number of events in the target interval; r: average events per unit time/space; t: number of such units; μ: mean; σ^2: variance. Assumes a constant event rate.

Exam rule. If the question says 4 calls *per hour*, then $\lambda = 4$ for 1 hour, $\lambda = 2$ for 30 minutes, and $\lambda = 12$ for 3 hours. Always adjust the rate to the requested interval.

Poisson approximation to binomial

$$\lambda = np, \qquad X \approx \mathrm{Poisson}(\lambda)$$

Symbols. n: independent binomial trials; p: small success chance; λ: expected successes. This is an approximation when n is large and p is small; use exact binomial if requested and feasible.

Geometric: trial of the first success

$$P(X = k) = (1-p)^{k-1} p, \quad k = 1, 2, \ldots$$

Symbols. X: *trial number* of first success; p: success probability on each independent trial; k: trial on which first success occurs. Then $E(X) = 1/p$, $\mathrm{Var}(X) = (1-p)/p^2$.

Geometric: wait beyond k trials

$$P(X > k) = (1-p)^k$$

Symbols. X: trial number of first success (starts at 1); k: completed failures; p: per-trial success probability. This means k trials have all failed.

Exam rule. Some textbooks define geometric Y as *failures before* the first success (starts at 0). Then $P(Y = k) = (1 - p)^k p$ and $Y = X - 1$. State the convention used.

Discrete uniform: all m values equally likely

$$P(X = x_i) = \frac{1}{m}$$

Symbols. X: random variable taking the distinct values $x_1, \ldots, x_m$; m: number of possible values; i: value index.

4.3 Discrete inequality rules: binomial, Poisson, geometric

Use these for any nonnegative integer count with PMF $p(k) = P(X = k)$.

Request	Calculate
$P(X < k)$	$p(0) + \cdots + p(k-1)$
$P(X \le k)$	$p(0) + \cdots + p(k)$
$P(X \ge k)$	$1 - [p(0) + \cdots + p(k-1)]$
$P(X > k)$	$1 - [p(0) + \cdots + p(k)]$
$P(a \le X \le b)$	$p(a) + p(a+1) + \cdots + p(b)$

Meaning. $p(j)$: probability of exactly j events; j: count being summed; a, b, k: integer cutoffs. If a count cannot be negative, start from 0.

Mini-example: Poisson with $\lambda = 3$.
$P(X < 2) = P(X = 0) + P(X = 1) = e^{-3}(1 + 3)$.
$P(X \ge 2) = 1 - e^{-3}(1 + 3)$.
$P(X > 2) = 1 - e^{-3}\left(1 + 3 + 3^2/2!\right)$.
"At least 2" includes 2; "more than 2" excludes 2.

4.4 Continuous distributions

A probability density function (PDF) $f_X(x)$ is not itself the probability at one point. For a continuous variable, probability is *area under the density* over an interval. Total area is 1.

Continuous uniform on $[a, b]$

$$f_X(x) = \frac{1}{b - a} \quad (a \le x \le b)$$

Symbols. X: continuous measurement; a: lower endpoint; b: upper endpoint, $b > a$; f_X: density. Density is zero outside $[a, b]$.

Uniform probability, mean and variance

$$P(c \le X \le d) = \frac{d - c}{b - a}, \quad \mu = \frac{a + b}{2}, \quad \sigma^2 = \frac{(b - a)^2}{12}$$

Symbols. $[a, b]$: full uniform interval; $[c, d]$: requested interval within it; μ: mean; σ^2: variance. Clip the requested interval to $[a, b]$ if necessary.

Exponential waiting-time density

$$f_X(x) = \lambda e^{-\lambda x} \quad (x \ge 0)$$

Symbols. X: waiting time; x: nonnegative duration; $\lambda > 0$: constant event rate per unit time; f_X: PDF. For $x < 0$, density is zero.

Exponential CDF and right tail

$$P(X \le t) = 1 - e^{-\lambda t}, \qquad P(X > t) = e^{-\lambda t}$$

Symbols. X: waiting time; $t \ge 0$: time cutoff; λ: rate. $E(X) = 1/\lambda$; $\mathrm{Var}(X) = 1/\lambda^2$.

Exponential between two times

$$P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$$

Symbols. $0 \le a < b$: interval boundaries; λ: event rate; X: waiting time. Strict and inclusive endpoints have equal probability.

Normal: change a measurement to z

$$Z = \frac{X - \mu}{\sigma}, \qquad z = \frac{x - \mu}{\sigma}$$

Symbols. X: normally distributed measurement; x: numerical cutoff; μ: population mean; $\sigma > 0$: population SD; Z: standard-normal variable; z: its value at cutoff x.

Normal table rule. Define $\Phi(z) = P(Z \le z)$, the area *to the left* of z, for $Z \sim N(0, 1)$.

Question	Left-cumulative-table calculation
Below a	$\Phi(z_a)$
Above a	$1 - \Phi(z_a)$
Between a, b	$\Phi(z_b) - \Phi(z_a)$
Top q fraction	Find z with $\Phi(z) = 1 - q$
Bottom q fraction	Find z with $\Phi(z) = q$

Meaning. $z_a = (a - \mu)/\sigma$, $z_b = (b - \mu)/\sigma$; $a < b$: original cutoffs; q: probability fraction (e.g., 0.05 for 5%). If the normal table gives area from 0 to z, adjust by 0.5 rather than treating it as a left-cumulative table.

Normal percentile: convert back to original units

$$x = \mu + z\sigma$$

Symbols. x: cutoff in original units; μ: mean; σ: SD; z: normal-table quantile having the desired *left-side* area.

Normal distribution notation

$$X \sim N(\mu, \sigma^2)$$

Symbols. X: normal random variable; μ: mean; σ^2: variance, *not* SD. If mean is 50 and SD 10, write $N(50, 100)$.

Optional normal approximation to binomial

$$\mu = np, \qquad \sigma = \sqrt{np(1-p)}$$

Symbols. n: trials; p: success chance; μ, σ: parameters of the approximating normal count. Check np and $n(1-p)$ are sufficiently large (often at least 5 or 10).

Exam rule. Normal approximation for integer counts requires a continuity correction: $P(X \le k)$ uses boundary $k + 0.5$; $P(X \ge k)$ uses $k - 0.5$. No such correction is needed for an originally continuous normal variable.

5. Unit 5 — Random variables, PMFs, PDFs and expectation

5.1 First decide whether the variable is discrete or continuous

Discrete X	Takes separate values, e.g., 0, 1, 2 defects. Use PMF $p_X(x) = P(X = x)$. Add probabilities.
Continuous X	Takes values along an interval, e.g., waiting time. Use density $f_X(x)$. Integrate to obtain probabilities.

PMF conditions and missing constant

$$p_X(x) \geq 0, \qquad \sum_x p_X(x) = 1$$

Symbols. $p_X(x)$: probability assigned to discrete value x; $\sum_x$: sum over all possible values. To find unknown c in $p_X(x) = cg(x)$, set $c \sum_x g(x) = 1$.

PDF conditions and missing constant

$$f_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

Symbols. f_X: continuous probability density; x: measurement. To find unknown c in $f_X(x) = cg(x)$ on $[a, b]$, set $c \int_a^b g(x)\,dx = 1$.

CDF definition: works for every random variable

$$F_X(x) = P(X \leq x)$$

Symbols. F_X: cumulative distribution function; X: random variable; x: cutoff. For a discrete variable sum PMF values up to x; for continuous variables integrate the density up to x.

Construct a CDF

$$F_X(x) = \sum_{u \leq x} p_X(u) \text{ (discrete)}, \quad F_X(x) = \int_{-\infty}^{x} f_X(u)\,du \text{ (continuous)}$$

Symbols. u: dummy value summed or integrated; x: upper cutoff; p_X: PMF; f_X: density. A valid CDF starts at 0, never decreases and approaches 1.

Interval probability from any CDF

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

Symbols. $a < b$: endpoints; F_X: cumulative probability. For *discrete* values, this formula excludes a and includes b. For continuous values, endpoints do not matter.

Expected value: discrete

$$E(X) = \sum_x x\, p_X(x)$$

Symbols. X: discrete random variable; x: possible value; $p_X(x)$: probability of that value; $E(X)$: expected value or long-run mean.

Expected value: continuous

$$E(X) = \int_{-\infty}^{\infty} x\, f_X(x)\,dx$$

Symbols. X: continuous random variable; x: possible value; f_X: PDF; $E(X)$: expected value, provided the integral exists.

Second moment and variance

$$E(X^2) = \sum_x x^2 p_X(x), \quad \text{Var}(X) = E(X^2) - [E(X)]^2$$

Symbols. X: discrete random variable in the first expression; p_X: PMF; $E(X^2)$: expected squared value; $\text{Var}(X)$: population variance. For continuous X, replace the sum by $\int x^2 f_X(x)\,dx$.

Standard deviation of a random variable

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Symbols. X: random variable; $\text{Var}(X)$: its variance; $\text{SD}(X)$: standard deviation in the same units as X.

Expected value of a function

$$E[g(X)] = \sum_x g(x) p_X(x)$$

Symbols. g: known function; X: discrete random variable; x: each possible input; p_X: PMF. For continuous X, use $\int g(x) f_X(x)\,dx$.

Shift and rescale a random variable

$$E(aX + b) = aE(X) + b, \quad \text{Var}(aX + b) = a^2 \,\text{Var}(X)$$

Symbols. a, b: fixed numbers; X: random variable; E: expectation; Var: variance. Adding b changes the mean but *not* the variance.

Distribution of a transformed discrete variable

$$P(Y = y) = \sum_{x:\, g(x)=y} P(X = x)$$

Symbols. $Y = g(X)$: transformed variable; g: transformation; y: specified output; sum probabilities of *every* input x producing output y.

Fast method for a PMF question

1. List each possible value of X, and its probability.
2. Check probabilities add to 1; solve any missing constant.
3. For the mean, make a column $x\,p(x)$ and sum it.
4. For the variance, make a column $x^2 p(x)$, sum it, and subtract $[E(X)]^2$.

Exam rule. A PDF may be greater than 1 at a point. Only its *area* over the full range must equal 1. Do not calculate $P(X = 3)$ from a continuous PDF: that probability is zero.

6. Unit 6 — Two-dimensional random variables

Core idea. A *joint* distribution describes two variables together. A *marginal* distribution describes one variable alone. A *conditional* distribution describes one variable when the other is known.

6.1 Discrete joint PMF and marginal probabilities

Joint PMF and total probability

$$p(x, y) = P(X = x, Y = y), \qquad \sum_x \sum_y p(x, y) = 1$$

Symbols. X, Y: discrete random variables; x, y: allowed values; $p(x, y)$: joint probability of that pair. Every entry must be

nonnegative.

Marginals: sum out the other variable

$$p_X(x) = \sum_y p(x, y), \qquad p_Y(y) = \sum_x p(x, y)$$

Symbols. $p(x, y)$: joint PMF; $p_X(x)$: probability that $X = x$; $p_Y(y)$: probability that $Y = y$; sums run over all possible values of the variable being removed.

Conditional PMF

$$p_{Y|X}(y \mid x) = \frac{p(x, y)}{p_X(x)}$$

Symbols. $p(x, y)$: joint probability; $p_X(x) > 0$: marginal probability of known $X = x$; $p_{Y|X}(y \mid x)$: probability of $Y = y$ given $X = x$.

Test whether two discrete variables are independent

$$p(x, y) = p_X(x)p_Y(y) \quad \text{for every } (x, y)$$

Symbols. $p(x, y)$: joint PMF; p_X, p_Y: marginal PMFs. Equality must hold for *all* possible pairs. One counterexample proves dependence.

Expected value of a product

$$E(XY) = \sum_x \sum_y xy\, p(x, y)$$

Symbols. X, Y: discrete variables; x, y: their values; $p(x, y)$: joint PMF; $E(XY)$: expected product.

Covariance between random variables

$$\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

Symbols. X, Y: random variables; $E(XY)$: mean product; $E(X), E(Y)$: separate means; Cov: population covariance.

Population correlation

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Symbols. ρ: population Pearson correlation; Cov: covariance; $\sigma_X, \sigma_Y > 0$: population SDs. $-1 \le \rho \le 1$.

Expected sums and variance of sums

$$E(X \pm Y) = E(X) \pm E(Y), \quad \mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{Cov}(X, Y)$$

Symbols. X, Y: variables with finite variances; E: expected value; Var: variance; Cov: covariance. For independent variables, covariance is zero. For a difference, use the minus sign before 2 Cov.

6.2 Continuous joint density

Joint density: total volume and event probability

$$\int\int f_{X,Y}(x,y)\,dx\,dy = 1, \quad P((X,Y) \in A) = \iint_A f_{X,Y}(x,y)\,dx\,dy$$

Symbols. $f_{X,Y}$: joint density, nonnegative; x, y: coordinates of the two measurements; A: region of interest; first integral is over the entire allowed region.

Marginal densities and independence

$$f_X(x) = \int f_{X,Y}(x,y)\,dy, \qquad f_{X,Y}(x,y) = f_X(x)f_Y(y) \text{ if independent}$$

Symbols. $f_{X,Y}$: joint density; f_X, f_Y: marginal densities; integrate out the variable not wanted, over its allowed range. Independence requires the product relation almost everywhere.

Conditional continuous density

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

Symbols. $f_{X,Y}$: joint density; $f_X(x) > 0$: marginal density at conditioning value x; $f_{Y|X}$: conditional density of Y.

How to solve a joint PMF table

1. Put every allowed (x, y) pair in a grid. If an unknown multiplier c is given, add all cell expressions and set the result equal to 1.
2. Add each row and column. Those totals are the two marginal distributions.
3. For a conditional probability, divide the matching cell by the marginal probability of the known value.
4. For independence, compare cell probabilities with products of row and column marginal probabilities.

Exam rule. A table with values only at $x, y = 0, 1, 2$ is a *joint PMF*, even if the question accidentally calls it a "density". Discrete values require sums, not integrals.

7. Unit 7 — Hypothesis testing and inference

7.1 Set up the hypothesis before choosing the statistic

Four required items

1. H_0: null hypothesis, usually equality to the claimed value.
2. H_1: alternative hypothesis, chosen from the question's wording.
3. α: significance level (e.g., 5% means $\alpha = 0.05$); confidence level is $1 - \alpha$.
4. Test statistic and matching critical value. Compare them; report *reject* or *do not reject* H_0.

Translate the alternative accurately

Question wording	Alternative H_1	Test
Different, changed, not equal	$\mu \neq \mu_0$	Two-tailed
Greater, increased, exceeds	$\mu > \mu_0$	Right-tailed
Smaller, decreased, below	$\mu < \mu_0$	Left-tailed

Meaning. μ: unknown population mean; μ_0: claimed or reference mean. The same direction rules apply to differences $\mu_1 - \mu_2$.

Exam rule. A question asking whether a mean is "the same" uses $H_1 : \mu \neq \mu_0$, which is *two-tailed.* A claim that a mean "has increased" uses $H_1 : \mu > \mu_0$, which is *right-tailed*.

7.2 Choose the correct test

Situation	Standard procedure
One mean; population SD σ known	One-sample z-test
One mean; population SD unknown	One-sample t-test, with normality or suitable large-sample conditions
Two independent means; known population SDs	Two-sample z-test
Two independent means; unknown SDs, no equal-variance assumption	Welch two-sample t-test
Two independent means; common population variance expressly assumed	Pooled two-sample t-test
Before/after readings on the *same* individuals	Paired t-test on within-person differences
Large samples; approximate normal method explicitly requested	Large-sample z approximation may be appropriate; state assumptions

Conditions. Observations should be selected and observed independently within the relevant samples. For small-sample t-tests, the underlying population or paired differences should be approximately normal. Separate treatment groups must be independent for two-independent-sample tests. Do not use an equal-variance pooled formula without that assumption.

7.3 Critical values: the entire decision rule

Critical-value notation
$t_{q,\nu}$ means the value with area q *to its left* under a t distribution with ν degrees of freedom.
z_q means the corresponding left-area quantile under the standard normal distribution.
For the table below, c is the *positive* critical magnitude; T is the calculated statistic, either t or z.

Rejection rules — no p-values

Alternative	Positive critical value c	Reject H_0 when
$H_1 : \mu > \mu_0$	$t_{1-\alpha,\nu}$ or $z_{1-\alpha}$	$T > c$
$H_1 : \mu < \mu_0$	$t_{1-\alpha,\nu}$ or $z_{1-\alpha}$	$T < -c$
$H_1 : \mu \neq \mu_0$	$t_{1-\alpha/2,\nu}$ or $z_{1-\alpha/2}$	$\lvert T\rvert > c$

Meaning. H_0: null; H_1: alternative; μ: true mean; μ_0: reference mean; α: significance level; ν: degrees of freedom; T: computed t or z; $c > 0$: critical cutoff. If exactly on the boundary, follow the course's stated rejection convention; using $\geq$ versus $>$ has no practical effect for continuous null distributions.

How to read your t-table

- *Cumulative/left-tail table:* for one tail use $1 - \alpha$; for two tails use $1 - \alpha/2$.
- *Upper-tail table:* for one tail use column α; for two tails use column $\alpha/2$.
- *Two-tail significance table:* for a two-tailed test use the column headed α, not $\alpha/2$ again.
- Always select the row for the correct degrees of freedom ν. A z critical value needs no degrees of freedom.

Common standard-normal critical magnitudes

α	Confidence $1 - \alpha$	One tail $z_{1-\alpha}$	Two tails $z_{1-\alpha/2}$
0.10	90%	1.282	1.645
0.05	95%	1.645	1.960
0.01	99%	2.326	2.576

Meaning. α: probability of a Type I error; critical magnitudes shown to three decimals. For t tests use the t-table, not these z values.

7.4 One-sample tests

One-sample t-test: unknown population SD

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \nu = n - 1$$

Symbols. $\bar{x}$: sample mean; μ_0: mean claimed by H_0; s: sample SD using divisor $n - 1$; n: sample size; ν: degrees of freedom; t: calculated statistic.

One-sample z-test: known population SD

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Symbols. $\bar{x}$: sample mean; μ_0: null mean; σ: known population SD; n: sample size; z: calculated statistic.

Standard error for one mean

$$\mathrm{SE}_{\bar{x}} = \frac{s}{\sqrt{n}} \quad \text{(estimated)}, \qquad \frac{\sigma}{\sqrt{n}} \quad \text{(known SD)}$$

Symbols. s: sample SD; σ: population SD; n: sample size; SE: standard deviation of the sampling distribution of the sample mean.

One-sample t-test mini-example
Claim: $\mu_0 = 50$. Sample: $n = 16$, $\bar{x} = 53$, $s = 6$. Test *whether it increased* at 5%.
$H_0 : \mu = 50,\ H_1 : \mu > 50$. Then $t = (53 - 50)/(6/\sqrt{16}) = 2.00$, $\nu = 15$.
One-tail critical value: $t_{0.95,15} \approx 1.753$. Since $2.00 > 1.753$, **reject** H_0. The data support an increase at 5%.

7.5 Two independent-sample tests

Standard error: two independent means

$$\mathrm{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Symbols. s_1, s_2: sample SDs; n_1, n_2: sample sizes; SE: estimated standard error of $\bar{x}_1 - \bar{x}_2$. This does *not* assume equal variances.

Welch two-sample t-test (unequal variances allowed)

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{\mathrm{SE}}$$

Symbols. $\bar{x}_1, \bar{x}_2$: group means; Δ_0: difference claimed by H_0, normally 0; $\mathrm{SE} = \sqrt{s_1^2/n_1 + s_2^2/n_2}$; t: test statistic.

Welch degrees of freedom (calculate only when required)

$$v_1 = \frac{s_1^2}{n_1}, \quad v_2 = \frac{s_2^2}{n_2}, \quad \nu \approx \frac{(v_1 + v_2)^2}{\dfrac{v_1^2}{n_1 - 1} + \dfrac{v_2^2}{n_2 - 1}}$$

Symbols. s_1, s_2: sample SDs; n_1, n_2: group sizes; v_1, v_2: contributions to squared standard error; ν: approximate t degrees of freedom. This is *not* automatically $n_1 + n_2 - 2$.

Pooled variance: equal population variances assumed

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Symbols. s_1^2, s_2^2: sample variances; n_1, n_2: sample sizes; s_p^2: pooled variance estimate. Use only if the two population variances are assumed equal.

Pooled two-sample t-test

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{s_p\sqrt{1/n_1 + 1/n_2}}, \qquad \nu = n_1 + n_2 - 2$$

Symbols. $\bar{x}_1, \bar{x}_2$: group means; Δ_0: null difference, usually 0; $s_p = \sqrt{s_p^2}$: pooled SD; n_1, n_2: group sizes; ν: degrees of freedom.

Two-sample z-test: known population SDs

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

Symbols. $\bar{x}_1, \bar{x}_2$: sample means; Δ_0: null mean difference; σ_1, σ_2: known population SDs; n_1, n_2: sample sizes.

Exam rule. Compare the same statistic with the matching distribution: Welch t uses Welch ν; pooled t uses $n_1 + n_2 - 2$; large-sample z uses normal cutoffs. The *same* estimated SE can appear in a Welch t-test and a large-sample z approximation, but the critical values differ.

7.6 Paired t-test: matched or before/after measurements

Paired t-test on differences

$$d_i = x_{i,\text{after}} - x_{i,\text{before}}, \quad t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}, \quad \nu = n - 1$$

Symbols. d_i: difference for pair i (state its direction); $\bar{d}$: mean of n differences; d_0: difference claimed under H_0, usually 0; s_d: sample SD of the differences; n: number of pairs.

Exam rule. For a paired test, calculate *one difference per person*. Do not treat the before and after columns as independent groups. The sign of each difference must follow your stated direction.

7.7 Confidence intervals: same critical-value logic

Confidence interval for one mean (unknown SD)

$$\bar{x} \pm t_{1-\alpha/2,\,n-1}\frac{s}{\sqrt{n}}$$

Symbols. $\bar{x}$: sample mean; s: sample SD; n: sample size; $1 - \alpha$: confidence level; t: positive t quantile with $n - 1$ degrees of freedom.

Confidence interval for one mean (known SD)

$$\bar{x} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Symbols. $\bar{x}$: sample mean; σ: known population SD; n: sample size; z: positive standard-normal quantile; $1 - \alpha$: confidence level.

Confidence interval: two means with equal variances

$$(\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2,\,\nu}\, s_p\sqrt{1/n_1 + 1/n_2}$$

Symbols. $\bar{x}_1, \bar{x}_2$: sample means; s_p: pooled SD; n_1, n_2: group sizes; $\nu = n_1 + n_2 - 2$; $1 - \alpha$: confidence level. Equal population variances assumed.

Confidence interval: two means with unequal variances

$$(\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2,\,\nu}\,\mathrm{SE}$$

Symbols. $\bar{x}_1, \bar{x}_2$: sample means; $\mathrm{SE} = \sqrt{s_1^2/n_1 + s_2^2/n_2}$; ν: Welch degrees of freedom; $1-\alpha$: confidence level.

Exam rule. For a *two-sided* test of $H_0 : \mu_1 - \mu_2 = 0$, the corresponding $100(1-\alpha)\%$ confidence interval leads to rejection when it *does not include 0*. A confidence interval does not prove an included value is correct.

7.8 Type I error, Type II error and correct conclusions

Decision outcomes

Your decision	H_0 actually true	H_0 actually false
Reject H_0	Type I error	Correct detection
Do not reject H_0	Correct non-rejection	Type II error

Type I error, Type II error and power

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}), \quad \beta = P(\text{do not reject } H_0 \mid H_1 \text{ true}), \quad \text{Power} = 1-\beta$$

Symbols. α: Type I error probability; β: Type II error probability at a specified alternative; H_0: null hypothesis; H_1: alternative.

Use exact conclusion wording. If the statistic enters the rejection region: "Reject H_0. There is sufficient evidence for [specific alternative] at the stated significance level." Otherwise: "Do not reject H_0. The sample does not give sufficient evidence for [specific alternative]." *Never write that non-rejection proves H_0.*

Last-minute revision: one-page decision sheet

1. Identify what the question asks
Counts → Binomial / Poisson / Geometric.
Measurements → Uniform / Exponential / Normal.
"Given" → Conditional probability.
"Given the result, which source?" → Bayes.
Paired observations → Covariance, correlation or joint PMF.
"Test a mean" → Set up H_0, H_1 and select z or t.

2. Discrete cutoffs you cannot afford to mix up
$X < k$: stop at $k - 1$. $X \leq k$: include k.
$X \geq k$: subtract the values below k from 1.
$X > k$: subtract values through k from 1.
For continuous variables, strict and inclusive endpoints give the same probability.

3. High-frequency formulas, in one place

$$\bar{x} = \frac{\sum f_i m_i}{N}, \quad \text{Median} = L + \frac{N/2 - C}{f}\,h, \quad \text{Mode} = L + \frac{d_1}{d_1 + d_2}\,h$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}, \quad \text{CV} = \frac{s}{\bar{x}}\,100\%, \quad r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

$$P(X = k)_{\text{Bin}} = \binom{n}{k} p^k (1-p)^{n-k}, \quad P(X = k)_{\text{Pois}} = \frac{e^{-\lambda}\lambda^k}{k!}$$

$$z = \frac{x - \mu}{\sigma}, \quad P(a < X < b) = \Phi(z_b) - \Phi(z_a)$$

Meaning. m_i: midpoint; f_i: class frequency; N: total count; L, C, f, h: class lower boundary, cumulative frequency before target class, target-class count, class width; d_1, d_2: modal frequency differences; x_i: observations; n: sample size/trials; s: sample SD; d_i: rank differences in the Spearman formula; k: count; p: binomial success chance; λ: Poisson event mean; μ, σ: normal mean/SD; Φ: left-cumulative normal probability. Each full definition appears in its unit.

4. t-test: six steps, always in this order

1. State H_0 and H_1: $\neq$ means two-tail; $>$ right-tail; $<$ left-tail.
2. Choose significance level α.
3. Calculate t and degrees of freedom ν.
4. Read the matching positive t critical magnitude c.
5. Reject when right-tail $t > c$, left-tail $t < -c$, or two-tail $|t| > c$.
6. Explain the decision in the wording of the original question.

5. Which denominator?
One sample: SE = $s/\sqrt{n}$, degrees of freedom $n - 1$.
Two independent samples: SE = $\sqrt{s_1^2/n_1 + s_2^2/n_2}$ for Welch.
Only if equal population variances are assumed: use pooled variance, with degrees of freedom $n_1 + n_2 - 2$.

Past-paper focus. These formulas support the types of questions appearing across the supplied 2022–2026 worked papers: grouped frequency measures, CV and skewness, Spearman correlation, Bayes, binomial/Poisson counts, normal tails and percentiles, joint distributions, confidence intervals, and critical-value mean tests. No past-paper occurrence guarantees a future question.